



Applications Development and Emerging Technologies

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Leveraging *Likert Scales* in Software Acceptability Testing Surveys

Introduction to Likert Scales and Software Acceptability Testing

- **Purpose of Software Acceptability Testing:** This process ensures that software meets user expectations, identifies areas for improvement, and enhances user satisfaction. Likert scales are a key tool in quantifying subjective experiences related to software.

Introduction to Likert Scales and Software Acceptability Testing

- **What is a Likert Scale?:** A Likert scale is a psychometric scale often used to assess attitudes, perceptions, and satisfaction levels. Respondents rate their level of agreement or satisfaction using a set of ordered responses, typically ranging from “Strongly Disagree” to “Strongly Agree” (5- or 7-point scales).

Introduction to Likert Scales and Software Acceptability Testing

- **Importance in Software Testing:** Likert scales are essential in software acceptability tests, helping to measure users' attitudes towards specific aspects of the software, such as ease of use, functionality, and user satisfaction.

Designing and Administering Likert Scale Surveys

Scale Construction

- **Selecting Statements:** Statements should cover diverse aspects of software usability. Include statements about ease of use, speed, aesthetics, and functionality.
- **Balance Positive and Negative Statements:** Including both positively and negatively worded items minimizes response bias, ensuring a more accurate reflection of user sentiment.

Designing and Administering Likert Scale Surveys

Scale Construction

- **Choosing the Number of Response Points:** A 5-point scale is common, but 7-point or even more granular scales can be used for greater precision, depending on the level of detail needed.
- **Response Labels:** Clearly defined response points such as “Strongly Disagree,” “Disagree,” “Neutral,” “Agree,” and “Strongly Agree” help ensure consistent interpretation of the scale by respondents.

Analyzing Likert Scale Data

- Mean and Central Tendency:
 - **Mean:** Calculate the average score for each item to assess the overall sentiment of respondents.
 - **Mode:** The most frequent response, which may highlight the dominant perception among users.
 - **Median:** Useful when the data is skewed, offering insights into the middle point of user responses.

Analyzing Likert Scale Data

- Read more:
 - <https://bit.ly/3Cn7OYE>
 - <https://bit.ly/3YWSMBG>

Analyzing Likert Scale Data

- **Group Comparison:** Analyze differences in responses across user groups (e.g., novice vs. experienced users) to identify varying levels of satisfaction or different perceptions of the software.
- **Data Visualization:** Use bar charts, pie charts, and radar charts to present data clearly, making it easier to interpret and share insights.
- **Interpretation:** Focus on statements with lower average scores as these indicate areas where users experience dissatisfaction, providing clear targets for software improvement.

Leveraging Technology Adoption Models in Software Testing

- Two key models frequently used to analyze user acceptance and adoption of software are the **Technology Acceptance Model (TAM)** and the **Unified Theory of Acceptance and Use of Technology (UTAUT)**.
- These models can be integrated with Likert scale data to assess user perceptions and acceptance more effectively.

Leveraging Technology Adoption Models in Software Testing

Technology Acceptance Model (TAM)

- **Origins and Purpose:** TAM, developed by Davis in 1989, was adapted from the Theory of Reasoned Action. It was designed specifically to understand technology adoption and explain why individuals accept or reject technology.
- **Core Constructs:** TAM focuses on two key factors:
 - **Perceived Usefulness:** The belief that technology will enhance job performance or productivity.
 - **Perceived Ease of Use:** The belief that the technology is easy to use, requiring minimal effort.
- **Application in Software Testing:** TAM helps software developers assess whether the perceived usefulness and ease of use align with user needs, providing actionable feedback for improvements.

Leveraging Technology Adoption Models in Software Testing

- **Technology Acceptance Model (TAM):** TAM focuses on two primary factors that influence user acceptance of technology:
 - **Perceived Ease of Use:** Users are more likely to adopt software if they perceive it as easy to use. A Likert scale question could be, “I find this software easy to use,” with responses ranging from “Strongly Disagree” to “Strongly Agree.”
 - **Perceived Usefulness:** Users will adopt software if they believe it will enhance their performance. A related statement could be, “This software improves my productivity,” rated similarly.
- **Example in Software Testing:** In a user acceptability survey for a new project management tool, a Likert scale item assessing ease of use might show a majority of users strongly agreeing with “The tool is easy to learn and use.” If the average score is low, it signals a need for interface improvements.

Leveraging Technology Adoption Models in Software Testing

Unified Theory of Acceptance and Use of Technology (UTAUT)

- **Development:** Introduced by Venkatesh et al. in 2003, UTAUT integrates multiple technology acceptance theories, aiming to offer a comprehensive model for understanding user adoption.
- Core Constructs:
 - **Performance Expectancy:** The belief that using technology will yield performance benefits.
 - **Effort Expectancy:** The ease associated with using the technology.
 - **Social Influence:** The degree to which users perceive that significant others encourage its use.
 - **Facilitating Conditions:** The availability of resources and support for technology use.
- **Relevance to Software Testing:** By examining UTAUT's constructs, testers can understand user motivations more fully, especially how social and environmental factors might impact software acceptance.

Leveraging Technology Adoption Models in Software Testing

- **Unified Theory of Acceptance and Use of Technology (UTAUT):** UTAUT incorporates four key constructs influencing technology adoption:
 - **Performance Expectancy:** The degree to which users believe the software will help them achieve their goals (e.g., “This software helps me complete tasks faster”).
 - **Effort Expectancy:** The ease with which users can use the software (e.g., “The software is user-friendly”).
 - **Social Influence:** The degree to which users feel influenced by others (e.g., “Most of my colleagues recommend using this software”).
 - **Facilitating Conditions:** The availability of resources to support software use (e.g., “I have the resources to effectively use this software”).

Leveraging Technology Adoption Models in Software Testing

- **Using Likert Scales in UTAUT:** Each of the UTAUT constructs can be assessed using Likert scale items. For example, “This software is easy to use” would measure Effort Expectancy, while “I believe this software will help me achieve my goals” would assess Performance Expectancy.

Leveraging Usability Frameworks in Software Testing

Post-Study System Usability Questionnaire (PSSUQ)

- **Background:** Developed by IBM, PSSUQ is a standardized usability questionnaire to measure user satisfaction with software following an interaction or task.
- Focus Areas:
 - **System Usefulness:** Evaluates whether the software fulfills user needs effectively.
 - **Information Quality:** Assesses the clarity and helpfulness of information provided by the software.
 - **Interface Quality:** Measures the intuitive nature and aesthetics of the interface.
- **Application with Likert Scales:** PSSUQ's structured Likert scale items enable easy analysis of these aspects, allowing testers to gather specific feedback on usability, which aids iterative software improvements.

Leveraging Usability Frameworks in Software Testing

Post-Study System Usability Questionnaire (PSSUQ)

- The **Post-Study System Usability Questionnaire (PSSUQ)** is a widely used tool to assess users' satisfaction with a system after interacting with it. It includes Likert scale questions to evaluate several aspects of usability:
- Example Questions:
 - **System Effectiveness:** "The system meets my needs." (5-point Likert scale: Strongly Disagree to Strongly Agree)
 - **System Learnability:** "I could quickly learn to use the system." (5-point Likert scale)
 - **Satisfaction:** "I am satisfied with the performance of this software." (5-point Likert scale)
- **Data Interpretation:** Higher average scores on items in the PSSUQ indicate better system usability, while lower scores may point to areas where improvements are necessary, such as system design or functionality.

Leveraging Usability Frameworks in Software Testing

System Usability Scale (SUS)

- **Overview:** Developed by John Brooke in 1986, SUS is a quick and reliable tool for assessing system usability across diverse products and services. The scale consists of 10 standardized questions, typically rated on a 5-point Likert scale, and yields a single score between 0 and 100.
- **Scope:** SUS assesses general usability, focusing on ease of use and user satisfaction without delving into specific components like functionality or interface quality.
- **Benefits:**
 - **Simplicity and Speed:** Due to its concise format, SUS is efficient, easy to administer, and provides quick insights.
 - **Wide Applicability:** SUS is commonly used for initial testing phases or rapid usability assessments.

Leveraging Usability Frameworks in Software Testing

Differences Between SUS and PSSUQ

Aspect	SUS	PSSUQ
Structure	10 items, general usability score	Multiple sections (System Usefulness, Information Quality, Interface Quality)
Focus	Overall usability, broad user satisfaction	Detailed usability feedback on specific areas
Application	Ideal for initial or high-level usability assessments	Suitable for in-depth usability analysis after task completion
Analysis Depth	Single score, limited insight into specifics	Allows detailed feedback per usability aspect

Leveraging Usability Frameworks in Software Testing

When to Use SUS vs. PSSUQ

- **Use SUS when:**

- You need a quick, high-level snapshot of usability.
- Testing a wide range of systems where in-depth analysis isn't required.
- Conducting early-phase testing or A/B testing where simple comparisons are sufficient.

- **Use PSSUQ when:**

- Detailed, multi-dimensional feedback is necessary, particularly after users complete tasks with a system.
- You need insight into specific usability components (e.g., system usefulness vs. interface quality).
- Testing systems with more complex interfaces where nuanced feedback will guide development.

Practical Example

- Consider a software testing scenario where a new mobile app is being evaluated:
- Likert Scale Questions:
 - “The app is easy to navigate” (measuring ease of use, associated with TAM’s Perceived Ease of Use).
 - “The app improves my efficiency” (measuring Perceived Usefulness, part of TAM).
 - “I would recommend this app to others” (measuring Social Influence, part of UTAUT).
- **Survey Results Interpretation:** If the average response to “The app is easy to navigate” is a 4.2 (Agree), this indicates a positive reception regarding usability. However, if “The app improves my efficiency” scores a 2.8 (Disagree), the app’s usefulness could be questioned, prompting developers to enhance its functionality.

Ponder on this!

- Integrating Likert scale surveys with technology adoption models (TAM, UTAUT) and usability assessments like PSSUQ and SUS provides a robust framework for understanding user attitudes, acceptance, and satisfaction with software.
- By carefully designing and analyzing these surveys, software developers can identify key areas for improvement and make data-driven decisions to enhance the user experience.